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Green Service Consumption: Unlocking Customer Expectations on Technological Transformations Enhancing Purchase Experience in Retail Store

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ABSTRACT

The research delves into the intricacies of green service consumption within the retail sector, concentrating specifically on delineating customer expectations and elucidating the transformative technological interface. The study elucidates the intricate dynamics of customer expectations and technological transformations to enhance the purchase experience in retail stores, focusing on the evolving paradigm in the retail landscape from a product-centric to a service-oriented economy and unveiling the pivotal role of societal influences, technology and customer expectations. The central theme revolves around understanding green service consumption within the retail sector, emphasizing the interplay between societal values, technological innovations and the dynamism of customer expectations. The research design employed is qualitative, leveraging a multidisciplinary approach and gleaning insights from sociotechnical perspectives, environmental sustainability, AI and consumer behavior. The sampling design involved experts in the field who participated in interviews and were purposefully selected based on their expertise in green service consumption and retail practices. The analysis utilized the fuzzy DEMANTEL method, involving a rudimentary exploration of dimensions. The research leverages extensive and in-depth deliberations involving theoretical explorations, connections among influencing factors and methodological insights from fuzzy logic and DEMATEL to dissect the complexities of green service consumption. The study elucidates the pivotal transition in the retail landscape, unveiling the changing nature of customer expectations, the surging significance of green service consumption and the influence of societal values and technological advancements. The discourse signifies the paramount role of societal values, particularly the increasing awareness of sustainability, ethical sourcing and eco-friendly strategies, in shaping retail strategies. The research presents a multifaceted view of green service consumption, thereby contributing to an enriched theoretical landscape of green service-focused retail experiences.

1. Introduction

In the contemporary global context, two groundbreaking trends are currently exerting their influence on the retail sector: an escalating commitment to sustainable methodologies and an unceasing acceleration of technological progress (Clark & York, 2005). A significant transformation has swept through the realm of customer expectations, primarily driven by the ongoing revolution in technological advancements (Ramakrishna et al., 2020; Wongmonta, 2021; Pal et al., 2023). Concurrently, a substantial increase in labor costs has imposed a pressing demand on the retail industry (Larue, 2020; Anner, 2020). This dynamic landscape compels retailers to undergo profound and comprehensive transformations. This imperative for change is not a mere matter of choice but an essential response to the shifting paradigms of the modern marketplace (Brinca et al., 2021). As environmental concerns take center stage in consumer awareness and technological innovations continue to redefine human interaction with the world, retailers stand at a decisive juncture, marked by unprecedented opportunity and profound challenge. Comprehending and effectively harnessing the connection between eco-friendly service consumption and technological transformation consisting of a combination of AI, IoT, automation, VR and AR emerges as the essential prerequisite for achieving success in the retail industry (Sarkar, 2013; Chin et al., 2022; Ghosh & Jhamb, 2022).

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Modern consumers display an increased inclination towards environmental mindfulness, relying more on technology to shape and enhance retail experiences (Ghosh et al., 2022) and strongly demanding products and services that align with ecological principles. Moreover, customers expect seamless technological interactions throughout their shopping journeys (Shalender & Yadav, 2019; Tandon et al., 2021; Nair et al., 2024). The convergence of sustainability and the desire for technological advancements has ushered in a new retail era, necessitating a comprehensive exploration of the complexities governing consumer expectations. Subsequently, the study undertakes a multifaceted inquiry to elucidate the intricate dynamics at play when customer expectations of environmentally conscious, technologically astute consumers step into retail establishments.

Fundamental questions regarding the interplay between eco-friendly service consumption and technological transformations are explored, seeking profound implications for the dynamics of the retail experience and demystifying customer expectations. The study seeks to cultivate actionable knowledge and implementable strategies to empower retail entities to navigate a future characterized by synergistic co-existence between sustainability practices and technological advancements. This, in turn, aims to encourage the growth of the retail industry ecosystem. A significant shift from a product-centric to a service-oriented model necessitates a profound understanding of customer expectations (Sharma et al., 2022). As retailers grapple with the challenge of meeting these evolving expectations, technology emerges as a transformative force shaping the retail sector (Dwivedi et al., 2021). From in-store and online integration to sustainability practices and engagement through various technological dimensions, the study aims to scrutinize the profound impact of technology on enhancing the overall customer experience. The exploration of these dimensions not only sheds light on the complex web of relationships among influencing factors but also underscores the imperative for retailers to adapt and innovate in response to dynamic customer preferences and market dynamics (Kar et al., 2021; Nair et al., 2023). Consequently, the study elucidates the interconnectedness of customer expectation, technology transformation and customer experience, providing valuable insights for retail management and contributing to the evolving discourse on green service-focused retail experiences.

The study is aware of the urgency of addressing these paramount concerns, as it is not only imperative for retail establishments' survival and prosperity but, more importantly, sustainability. To fully understand how people use green services, we want to find out what hidden opportunities exist when environmental awareness and technological progress come together. This approach would foster a retail environment that prioritizes customer satisfaction and promotes environmental sustainability (Shahade et al., 2023). The research delves further into the complexities of these transformative dynamics, providing guidance for retailers seeking to thrive in an era where sustainability and technology are the driving forces of consumer expectations and retail evolution. The United Nations articulation in Sustainable Development Goal 9 (SDG 9), which calls for the construction of resilient infrastructure, the promotion of inclusive and sustainable industrialization, and the fostering of innovation, further underscores the need (Ospina-Forero et al., 2022). For a sustainable and progressive economy, the concept of green service consumption emerges as a central pivot. Green service consumption represents an intricate and multifaceted process that operates within traditional retail stores (Ostrom et al., 2021). A multitude of interconnected factors influence its non-linear progression, gradual evolution, and inherent complexity.

Understanding and identifying these pivotal dimensions within this transformation process has taken center stage. A review of literature on the dimensions has been meticulously examined from two primary perspectives: agency and contextual context. When viewed from the agency perspective, scholars have dedicated substantial attention to discerning the influence of various actors, including customers, employees, suppliers and other participating entities, on green service consumption. Customers, as highlighted by Evanschitzky et al. (2020), stand as the focal point of retail operations. Similarly, customer engagement and employee participation in green service consumption have proven to be significant factors (Piligrimienė et al., 2020; Nair et al., 2023). Employees who actively participate in these initiatives are better able to understand customer needs, experience less uncertainty, and, in turn, contribute to the more effective design and development of innovative services.

Shifting focus to the contextual domain, scholarly endeavors have concentrated on investigating the impact of both internal and external resources and their respective capabilities on green service consumption (Tuan, 2021; Jiang et al., 2023). Furthermore, the influential role of technology in shaping green service consumption has been substantiated by the work of Luan et al., 2023. Notably, Bansal et al. (2023) has illuminated the role of an innovative organizational culture in expediting the transformation process across diverse retail establishments. Subsequently, the intricate and multifaceted landscape of customer expectations, technological transformations within traditional retail stores presents a multifaceted and dynamic field of inquiry, one that navigates comprehensively and systematically. Therefore, notwithstanding the recent research strides made within the existing literature regarding dimensions of customer expectations, it remains undeniable that numerous gaps persist. These gaps encompass several dimensions, including: (1) The scholarly community has, indeed, put forth various dimensions from their respective vantage points, yet the scholarly body of work remains fragmented, lacking an effective integration of dimensions of customer expectation and a framework connecting these to service provider and service receiver, (2) The current body of literature falls short in its ability to explore the dimensions to enhance technological transformation. These dimensions collectively constitute a vast, interwoven system, where interdependencies abound. Existing research have addressed these as discrete entities, overlooking the intricate causal tapestry that underlies their relationships. (3) A noticeable absence lies in the failure to identify the essential dimensions of customer expectation. While existing literature has diligently employed case studies, empirical analyses and various methodological tools, it has overlooked the isolation of the most significant dimensions through multifactorial interactions. Therefore, the research questions for the study are:

RQ 1. *"What are the key parameter's that influence customer expectations in the context of Technological Transformations Enhancing the Purchase Experience in Retail Stores, with a specific focus on Green Service Consumption?"*

RQ 2. *"How do the dimensions of the parameters interact to shape consumer preferences and satisfaction in the retail environment?"*

The study commences with the construction of a framework, harnessing insights garnered from an exhaustive review of the existing literature. Subsequently, this framework is further refined through the judicious incorporation of insights obtained from interviews conducted within the retail domain. Lastly, these empirically grounded insights are juxtaposed with the perspectives of domain experts, culminating in the discernment of a concise set of dimensions. The concluding phase of the research involves a dissection of the research findings, wherein the implications for management are expounded upon and the contributions of this research to the existing body of knowledge are elucidated.

2. Review of Literature

2.1. Customer Experience and Enhancement through Technological progress

The concept of customer experience is receiving significant scholarly attention as businesses recognize its crucial role in influencing customer satisfaction, fostering loyalty, and achieving overall business success

(Eckhardt et al., 2019). The paradigmatic shift lies in the increasingly influential role of technological advancements in molding and refining the customer journey. Fortmann et al. (2019) and "Bryan" Jean et al., 2014 emphasize the transformative influence of cohesive omnichannel experiences, facilitated by technological amalgamation. When customers anticipate a seamless transition across diverse channels, technological facets like mobile applications and responsive websites play an indispensable role in furnishing a unified customer experience. Artificial intelligence (AI) emerges as a pivotal agent in personalizing customer experiences. As elucidated by Joung & Kim, 2023, AI, particularly machine learning algorithms, enables businesses to scrutinize large datasets in order to discern individual customer preferences. This results in a more personalized approach to customer interactions, customizing recommendations, content, and services to meet the individual needs and expectations of each customer. Chatbots and virtual assistants have become integral components of strategic customer service implementations. According to Xue et al., 2020, these technological components significantly contribute to increased customer satisfaction by providing instantaneous responses to inquiries, thereby enhancing overall efficiency and responsiveness in customer engagements. The immediacy and ubiquity of these automated systems introduce an additional layer of convenience into the customer experience, where data analytics stands as a keystone for comprehending and enhancing customer experiences.

Rosário and Dias (2023), in their scholarly contribution, underscore the importance of customer journey mapping, underpinned by data analytics, in identifying avenues for enhancement. This strategic methodology ensures that organizations are judiciously informed when effecting changes to optimize the overarching customer experience. In parallel with data analytics, the deployment of mechanisms for customer feedback and sentiment analysis tools has become increasingly prevalent. According to Zhang et al., 2020 and Sudirjo et al. (2023), real-time feedback mechanisms and sentiment analysis confer upon businesses a nuanced comprehension of customer emotions and satisfaction levels. These insights empower organizations to make judicious decisions and expeditiously address customer needs and concerns. Nevertheless, the assimilation of technology in the pursuit of refined customer experiences is not bereft of challenges. Ethical considerations and apprehensions surrounding data privacy have ascended to the forefront. To cultivate and sustain trust, businesses must navigate a delicate balance between the advantages of personalized experiences and the conscientious and ethical use of customer data, as expounded by Cain (2023). Trunfio et al., 2022 research probes the potential impact of AR and VR in crafting immersive brand interactions, thereby enhancing customer engagement and satisfaction. Consequently, the confluence of customer experience and technological progress encapsulates a dynamic and evolving terrain.

2.2. Customer Expectations on Technological Transformations

Customer expectations assume a pivotal role in the determination of retail business success, as the expectations are intricately interwoven with the technological metamorphoses witnessed within the retail sector. Notably, Parasuraman et al.'s (1991) SERVQUAL model delineated five dimensions that exert a profound influence on the expectations of service quality, furnishing a foundational framework to comprehend how customers evaluate their interactions with retailers. Furthermore, the scholarly discourse has been enriched by the insights of Zhang et al. (2021) and Wolf and Steul-Fischer (2022), who expounded upon the determinants of customer expectations, accentuating the pivotal roles played by price, assortment, convenience and service. Their discernment underscored that customer expectations are underpinned by a dual influence, encompassing utilitarian facets such as price and assortment alongside hedonic dimensions like service quality and the ambiance of the retail environment. This distinction assumes heightened significance when contemplating the impact of technological transformations on these expectations. The proliferation of e-commerce, mobile applications and in-store technologies has forged novel avenues for enhancing the customer experience. For instance, the integration of mobile applications and websites has empowered customers to peruse products, compare prices and access product information at their leisure (Huang et al., 2023). This convenience factor assumes a pronounced role in shaping customers expectations, fostering a desire for seamless, omnichannel experiences.

Notwithstanding these challenges, technological transformations present a tapestry of opportunities for retailers to align with customer expectations. The notion of "smart stores" seamlessly integrates technology to elevate the in-store experience. Technologies like augmented reality (AR) and virtual reality (VR) empower customers to visualize products before committing to a purchase, thereby bolstering their confidence and overall satisfaction (Cranmer et al., 2021). Moreover, the integration of Internet of Things (IoT) devices in retail enables real-time insights into customer behavior, leading to personalized offers and enhanced inventory management (Hossain et al., 2021; Nair et al., 2024). Subsequently, retailers are compelled to adapt to these changes by delivering omnichannel experiences, ensuring the impregnability of customer data and harnessing innovative technologies to meet and surpass expectations.

Parameter 1. *In-Store and Online Integration*

One of the quintessential facets of contemporary retail lies in the provision of an unswerving omni-channel experience to patrons, accentuating the profound importance of amalgamating physical in-store and digital online shopping realms (Xie et al., 2023). A consistent omni-channel experience not only facilitates customers in transitioning effortlessly between tangible and virtual touchpoints but also encompasses services such as "Click and Collect" and the fusion of mobile applications with e-commerce platforms (Melkonyan et al., 2020). The identified quandary revolves around the intricacy of furnishing a seamless experience across these diverse channels, which could potentially lead to customer perplexity and discontentment (Sheu, 2011). The metrics and indicators such as net promoter score, customer effort score, customer satisfaction score, channel-specific conversion rates, omni-channel customer journey analysis, customer feedback and reviews, time to resolution and customer retention rates provide a clear and concise overview of the key aspects to consider when measuring customer satisfaction in the experience between online and offline channels (Boonstra, 2021). Customer satisfaction is the most important indicator of customer experience, demonstrating the ease with which customers can navigate the realms of online and offline shopping. To ameliorate this situation, retailers need to deploy robust software systems that consolidate inventories, prices and loyalty programs, thereby furnishing a unified customer experience (Cheshin et al., 2018).

Parameter 2. *Sustainability and Green Practices*

The contemporary consumer landscape is profoundly preoccupied with the tenets of sustainability and ecological responsibility (Garcia-Vazquez et al., 2021). This parameter encompasses the dissemination of comprehensive sustainable product information, the use of eco-friendly packaging materials, and the provision of transparent details regarding ethical sourcing practices. It underscores the growing emphasis on products that adhere to environmentally responsible and ethically sourced standards (Chong & Patwa, 2023). Transparently communicating these practices to customers is crucial for fostering trust and fostering customer loyalty. The current conundrum pertains to the provision of clear and comprehensive sustainable product information, eco-friendly packaging, and ethical sourcing details (Jarvenpaa & Essén, 2023). The primary customer experience indicator revolves around the level of trust and loyalty that customers place in the retailer, which is contingent on the perceptibility and accuracy of sustainable practices, packaging choices, and ethical sourcing information. To rectify this situation, retailers need to transparently articulate sustainability

endeavors, embrace the labeling of eco-friendly products and embark on educating customers on the nuances of these practices (Lukas & Welling, 2014).

Parameter 3. *Engagement and Interactivity*

Augmenting customer engagement through the prism of interactivity stands as a hallmark of contemporary retail paradigms. This parameter encompasses a diverse array of interactive experiences, inclusive of immersive virtual reality (VR) encounters, interfaces governed by gesture controls, intelligent mirrors, augmented reality (AR) product demonstrations, interactive store navigation aids, AR-furnished fitting rooms, real-time chat support, voice-activated assistance, interactive information kiosks and platforms fostering community engagement (Peddie, 2023). These technological innovations aspire to immerse customers within the shopping experience, rendering it more interactive and personalised. The entrenched dilemma resides in catering to a manifold of customer preferences and ensuring technological accessibility. The linchpin for assessing customer experience hinges upon the degree of engagement and contentment derived from these interactive experiences. To rectify this, retailers need to furnish an array of interactive experiences that are in harmony with distinct customer inclinations and enhance the accessibility and user-friendliness of the technology (Cuomo et al., 2020).

Parameter 4. *Transparency and Information*

Transparency and unimpeded access to information represent indispensable pillars for the cultivation of trust and assurance within the retail sphere (McGrath et al., 2021). This parameter pivots around the accessibility and transparency of both product information and pricing structures. Its raison d'être lies in guaranteeing that customers can readily access detailed product information and fathom the intricacies of pricing, thereby arming them with the insights to make judicious purchase decisions. The identified complication revolves around the provision of exhaustive product details and pricing transparency. The pivotal customer experience indicator crystallizes in the ease with which customers can avail themselves of comprehensive product information and perceive the transparency of pricing. To rectify this, retailers need to ascertain the veracity and accessibility of product information and institute transparent pricing frameworks that extend across multiple channels (Khandpur et al., 2020). Therefore, these four parameters are central in shaping customer expectations within the ambit of green service consumption and the integration of technology to enhance the retail store's purchase experience.

3. Proposed Conceptual Framework

The proposed conceptual framework (Fig. 1) elucidates the intricate process wherein a service provider furnishes both tangible and intangible services, concurrently enhancing service productivity through various parameters. The overarching process begins with a comprehensive overview, delineating the stages involved in the service provider's delivery of both tangible and intangible services. This

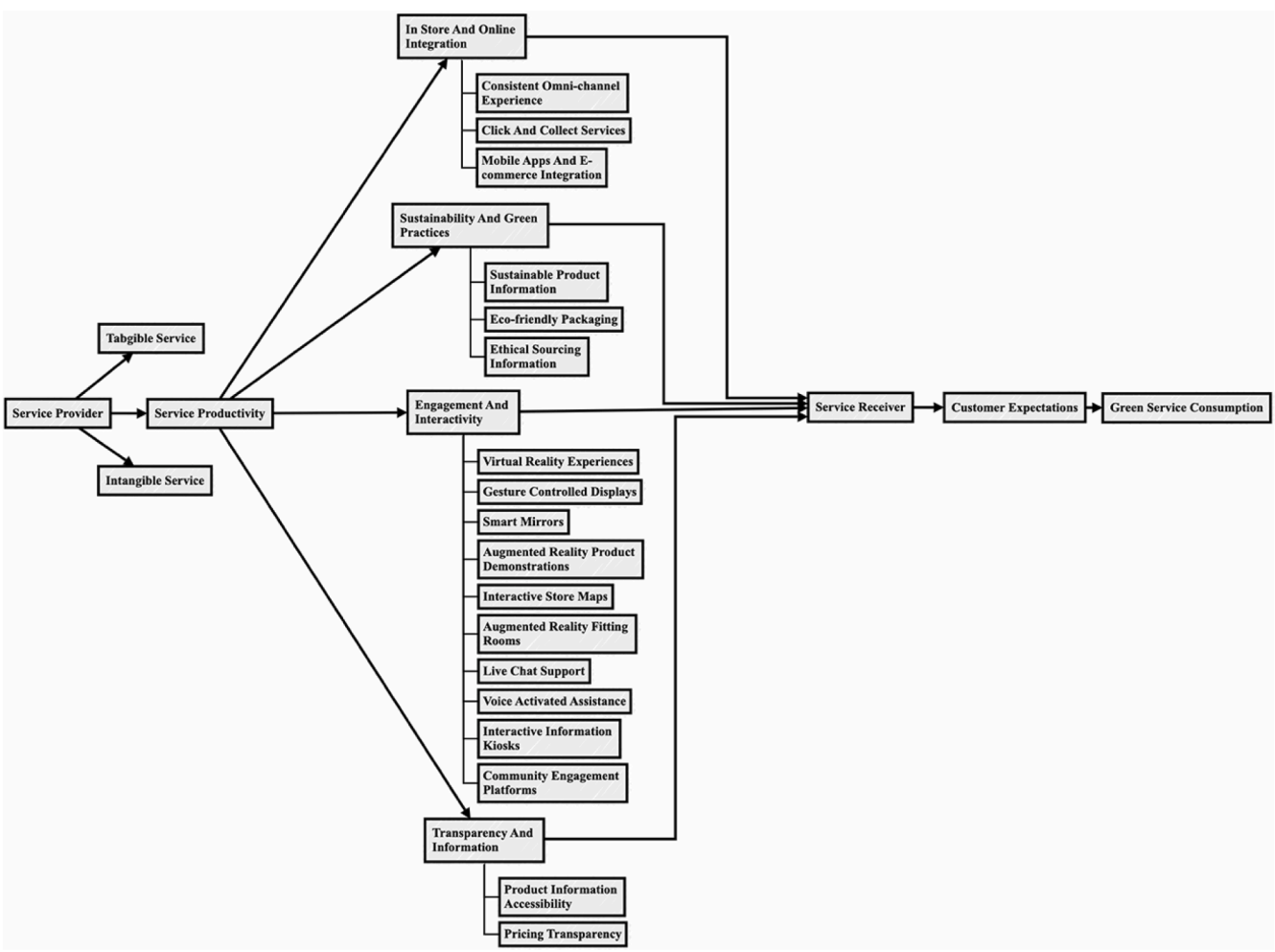

Fig. 1. Proposed Conceptual Framework of the process and dimensions Customer Expectations on Technological Transformations Enhancing Purchase Experience in Retail Store.

encompasses various aspects, including pre-purchase engagement, the actual purchase process and post-purchase interactions. Research has shown that a holistic view of the customer journey is essential for enhancing service productivity and customer satisfaction (Lemon & Verhoef, 2016). Mapping the entire customer journey, service providers can identify key touchpoints and areas for improvement, ensuring a seamless and satisfying customer experience. One significant parameter focuses on in-store and online integration, emphasizing the seamless blending of in-store and online experiences. This approach is supported by literature that highlights the importance of omnichannel strategies in retail (Jocevski et al., 2019). Integrating physical and digital channels allows retailers to offer a consistent and convenient shopping experience, which is increasingly demanded by consumers in the digital age. Addressing the escalating importance of sustainability, another parameter concentrates on sustainability and green practices. This parameter encompasses dimensions such as providing sustainable product information, implementing eco-friendly packaging practices and communicating ethical sourcing information.

Studies have demonstrated that consumers are increasingly concerned about environmental issues and prefer to support businesses that demonstrate sustainable practices (Young et al., 2010; Choi & Ng, 2011). Integrating sustainability into their operations, retailers can attract environmentally conscious consumers and differentiate themselves in the market.

The engagement and interactivity parameter underscores the significance of interactive elements, encompassing dimensions like virtual reality experiences, gesture-controlled displays and augmented reality product demonstrations. Literature suggests that interactive technologies can enhance customer engagement and create memorable shopping experiences (Bonfanti et al., 2023). Incorporating these technologies, retailers can provide immersive and personalized experiences that resonate with modern consumers. Transparency and information constitute another parameter, emphasizing clear communication. Dimensions within this parameter include ensuring product information accessibility and maintaining pricing transparency. For instance, sustainable practices in the framework can be intertwined with both in-store and online integration as well as engagement strategies. Acknowledging the dynamic nature of customer expectations, the framework considers the evolution of expectations in response to advancements in technological parameters and changing market dynamics. This dynamic perspective underscores the importance of adaptability and responsiveness in retail strategies. Studies have highlighted the need for businesses to be flexible and responsive to changes in consumer behavior and technological trends (Kristiana et al., 2024). Being adaptable, retailers can stay relevant and competitive in a rapidly evolving market. Therefore, the proposed conceptual framework offers a structured guide for service providers in the retail sector, helping them navigate the complex landscape of customer expectations, technological advancements, and sustainability. Integrating insights from relevant literature, the framework provides a comprehensive approach to enhancing service productivity and customer satisfaction. It emphasizes the importance of a holistic view of the customer journey, omnichannel integration, sustainability, engagement, transparency, and adaptability key elements that are critical for success in the modern retail environment.

4. Methodology

The underlying framework is predicated on the dimensions, aligns with the procedural steps outlined for the constructed Fuzzy DEMATEL method. Constructive feedback from experts participated pointed out a lack of convenience when juxtaposing the codes and corresponding names of individual dimensions. In response, the study augments the survey completion procedure by introducing a questionnaire reminiscent of the Likert scale. The collection of these questionnaires was facilitated through multiple channels, encompassing email, interviews and surveys. Considering that customer expectations transcend the confines of a single organization, the responses provided by employees less conversant with the intricacies of this multifaceted activity might yield results of questionable reliability. As such, the aim of this study is to administer the survey to middle and senior managers well-versed in the nuances of enhancing the purchase experience within the retail sector, along with academic expert's research niche lies within the domain. The majority of experts identified were actively engaged in the operations of retail stores. Among the experts from the academic realm, the selection predominantly comprised scholars renowned for their publications in authoritative journals. Additionally, these scholars boasted a track record of practical involvement in customer expectations within the retail sector and offered consultative services. This curation of experts ensured the high reliability of the sample data. In total, 51 questionnaires were distributed during the course of this study, yielding 40 responses. The final count of valid questionnaires that underwent rigorous scrutiny and analysis amounted to 28.

The raw data, initially procured through expert inquiries, underwent a transformative process involving triangular fuzzy conversion. This conversion paved the way for the calculation of the initial direct impact matrix. While the paper exclusively presents the original data from one of the experts, it's imperative to highlight the inherent conversion correlation between the fuzzy number, as outlined in Table 1 and the semantic evaluation of experts. Through adept programming in Microsoft Excel, the semantic evaluations provided by the experts were methodically translated into triangular fuzzy numbers. Subsequently, the Complex Fuzzy Cluster System (CFCS) method, guided by formulae (2) to (10), was meticulously employed to ascertain the direct influence matrix between the initial influencing factors. This process culminated in the tabulated presentation of the direct influence matrix, retaining three decimal places for precision. The present research encompasses a comprehensive exploration involving a systematic literature review and an in-depth interview analysis, which serves to elucidate the concept at hand and discern the dimensions implicated. Nevertheless, within the myriad factors under consideration, the precise assessment of individual factors significance and their intricate causal relationships remains a challenging endeavor. A notable gap in this domain pertains to the absence of quantitative substantiation for comprehending the pivotal factors. Subsequently, the paper employs the Fuzzy Decision-Making Trial and Evaluation Laboratory (DEMATEL) method to undertake a meticulously structured quantitative analysis (Bhalaji et al., 2019). Subsequently, the initial process of the Fuzzy DEMATEL method is executed, with the primary objective of identifying the issues and dimensions under scrutiny.

The section also incorporates a discussion of the data collection process, specifically concerning the aggregation of indicators illustrating the degree of influence, the degree of impact, the centrality degree and the causative degree for each dimension. The DEMATEL method, which stands for the Decision Test and Experimental Evaluation Method, constitutes a system factor analysis methodology primarily intended for addressing complex problems characterized by inconspicuous relationships. This approach relies on principles drawn from graph theory and matrix tools, harnessing the expertise and collective knowledge of experts and expert groups. It aspires to establish a visual framework predicated on the causal relationships that exist between diverse dimensions. The method, as previously evidenced by Du and Li (2021),

Table 1
TFN and Expert semantic evaluation.

Linguistics assessment	Corresponding score	Triangular fuzzy numbers
Negligible impact	0	(0,0,0.25)
Minimal impact	1	(0,0.25,0.5)
Limited influence	2	(0.25,0.5,0.75)
Substantial influence	3	(0.5,0.75,1)
Significant influence	4	(0.75,1,1)

involves the calculation of the centrality and causative degrees of these dimensions, culminating in the construction of a causal diagram. Dimensions are subsequently categorized as either causative or resultant, paving the way for the identification of key dimensions conducive to problem resolution, in alignment with the insights of (Santamaria et al., 2021) and (Salmen & Festing, 2022). The rationale behind opting for the Fuzzy DEMATEL method over a conventional DEMATEL approach lies in its adeptness at navigating the pervasive uncertainties and subjective elements inherent in real-world scenarios. Unlike its traditional counterparts, the Fuzzy DEMATEL framework possesses the unique capability to accommodate the inherent vagueness inherent in intricate relationships by permitting experts to articulate their assessments using fuzzy terms, thereby mirroring the nuanced nature of their analyses. This adaptability proves pivotal in encapsulating a wide array of viewpoints and interpretations, thereby fostering a more exhaustive examination.

Furthermore, the method's reliance on triangular fuzzy numbers serves to heighten the precision of the initial direct influence matrix, effectively capturing the subtleties embedded within expert evaluations. Despite encountering obstacles presented by the triangular fuzzy number format in direct matrix operations, the method adeptly employs defuzzification methodologies to elucidate outcomes, thereby enriching their applicability to decision-making endeavors. Hence, the Fuzzy DEMATEL approach presents itself as a robust mechanism for scrutinizing complex issues, embracing uncertainty, accommodating subjectivity, refining influence matrices and amplifying interpretative clarity, thereby solidifying its status as the preferred analytical tool for the present research pursuit. Consequently, decisions often assume non-quantifiable values and are expressed using subjective terms such as "better" and "satisfied." To tackle the challenge of fuzzy expert evaluations, this paper adopts the triangular fuzzy number method to refine the initial direct influence matrix, thereby enhancing the accuracy of the DEMATEL methodology. Notably, the triangular fuzzy number format is ill-suited for direct matrix operations and necessitates defuzzification, transforming it into clear values. To this end, a new direct relationship matrix need to be meticulously constructed. Among the array of defuzzification methods, this paper opts for the CFCS defuzzification method, as pioneered by Opricovic and Tzeng (2003) as a means to obtain clear and interpretable BNP values. The basic steps to determine the fuzzy DEMATEL are as follows:

Phase 1: In accordance with the research query, identify the factors exerting influence within the system, denoting them as D1, D2, D3 and so forth.

Phase 2: Construct a semantic assessment framework, partitioning the degree of influence amongst the various factors into five discernible tiers: insignificance denoted as "0," exceedingly feeble influence represented as "1," mild influence signified as "2," robust influence designated as "3," and exceptionally potent influence labeled as "4."

Phase 3: Procure the preliminary direct impact matrix. Solicit expertise in appraising the direct relationships between the factors, guided by their comprehension of the semantic evaluation chart. Consequently, generate a direct impact matrix in a sequential fashion.

Phase 4: Transform the semantic expressions into corresponding triangular fuzzy numbers using the semantic transformation reference. The triangular fuzzy number is a triplet, where "l" embodies a conservative value, the one closest to the factual value and "r" signifies the most optimistic value. The membership function of the triangular fuzzy number adheres to the framework depicted in Formula (1).

$$\mu_N(x) = \begin{cases} 0, & x < l \\ \dfrac{x-l}{m-l}, & l \le x \le m \\ \dfrac{r-x}{r-m}, & m \le x \le r \\ 0, & x > r \end{cases} \tag{1}$$

Phase 5: Employ the Complex System Cluster Fuzzy (CSCF) technique to de-fuzzify the fuzzy numbers, culminating in the calculation of the weighted mean based on the left and right ratings of the membership function to derive the comprehensive score. Each population score forms a novel initial direct impact matrix, delineated through the ensuing detailed measures.

(1) Normalizing TFN

$$xl_{ij}^k = \frac{l_{ij}^k - \min l_{ij}^k}{\Delta_{\min}^{\max}} \tag{2}$$

$$Xm_{ij}^k = \frac{m_{ij}^k - \min l_{ij}^k}{\Delta_{\min}^{\max}} \tag{3}$$

$$xr_{ij}^k = \frac{r_{ij}^k - \min l_{ij}^k}{\Delta_{\min}^{\max}} \tag{4}$$

$$\Delta_{\min}^{\max} = \max xr_{ij}^k - \min l_{ij}^k \tag{5}$$

(2) Calculate the ls and rs normalized values:

$$xls_{ij}^k = \frac{xm_{ij}^k}{\left(1 + xm_{ij}^k - xl_{ij}^k\right)} \tag{6}$$

$$xrs_{ij}^k = \frac{xr_{ij}^k}{\left(1 + xr_{ij}^k - xm_{ij}^k\right)} \tag{7}$$

(3) Calculating overall standard clear value:

$$x_{ij}^k = \frac{xls_{ij}^k * \left(1 - xls_{ij}^k\right) + xrs_{ij}^k * xrs_{ij}^k}{\left(1 - xl_{ij}^k + xrs_{ij}^k\right)} \tag{8}$$

(4) Calculating the clear value of the defuzzification of the expert k:

$$z_{ij}^k = \min l_{ij}^k + x_{ij}^k * \Delta_{\min}^{\max} \tag{9}$$

(5) Incorporate the unequivocal worth, signifying a comprehensive assessment by all experts and procure the quantitative value signifying the extent of direct influence of index factor i on j:

$$z_{ij} = \frac{Z_{ij}^1 + Z_{ij}^2 + \ldots + Z_{ij}^p}{p} \tag{10}$$

(6) The primary direct impact matrix is subject to manipulation for the acquisition of a standardized direct impact matrix $X = \left[X_{ij}\right]_{n*n}$ and $0 \le x_{ij} \le 1$. The calculation is as follows:

$$X = s * Z \tag{11}$$

$$s = \frac{1}{\max\limits_{1 \le i \le n} \sum_{j=1}^{m} z_{ij}}, \; i, j = 1, 2, \ldots, n \tag{12}$$

(7) Calculating the impact matrix $T = [t_{ij}]_{n*n}$.

The element denoted as t_{ij} encapsulates the implicit influence connection existing between factors i and j. The comprehensive influence matrix T embodies the global influence rapport between these elements. The calculation procedure for this matrix is explicated within equation (13).

$$T = \lim_{m\to\infty}\left(X + X^2 + X^3 + \ldots X^m\right) = X * (1 - X)^{-1} \tag{13}$$

(8) The influence degree, denoted as D_i, represents the extent of comprehensive impact that various factors exert on one another within the system. Conversely, the affected degree, indicated as R_j, quantifies the degree to which each factor is influenced by other factors in the system. The centrality degree is the summation of D_i and R_j, serving as a measure of the significance of factors within the system. Meanwhile, the cause degree, expressed as D_i- R_j, determines whether a factor falls into the cause group (in the event of a positive D_i - R_j) or the effect group (in the case of a negative D_i - R_j). The computation of these degrees is executed as follows:

$$D_i = \sum_{j=1}^{n} t_{ij} (\text{Influence Degree}) \tag{14}$$

$$R_j = \sum_{i=1}^{n} t_{ij} (\text{Affected Degree}) \tag{15}$$

$$P_{i=\{D_i+R_j|\ i=j\}} (\text{Center Degree}) \tag{16}$$

$$E_{i=\{D_i-R_j|\ i=j\}} (\text{Cause Degree}) \tag{17}$$

5. Analysis and Results

The validated data undergoes rigorous statistical processing, involving the application of the triangular fuzzy number method and defuzzification method, which culminate in the derivation of a direct influence matrix between the various dimensions. Leveraging the dataset guided by Formulae (11) through (13), the comprehensive impact matrix is diligently formulated, encapsulating the intricate web of dimensions that influence the parameters of customer expectation. Formulae (14) and (15) are then summoned into action, facilitating the computation of the influence degree, affected degree, centrality and cause degree for each individual dimension. These numerical values are systematically sorted. The causal distribution of each dimension, predicated on the cause degree, is artfully portrayed in a visual representation on the coordinate system. This manifests as a causal relationship diagram that serves to delineate the intricate interplay of these dimensions, as vividly illustrated in Fig. 2. However, whilst and Fig. 2 offer invaluable insights into the key dimensions wielding influence, the nuanced roles of these dimensions remain obfuscated in the cause-effect relationship diagram. This results in a lack of clarity concerning the interrelationships between these key dimensions. As articulated by Prendergast (2019), the setting of the threshold is a facet of this process, as it necessitates striking a judicious balance. If the threshold is established too low, it engenders a situation where the relationships between dimensions become excessively intricate and fail to provide the requisite information for decision-making. Conversely, a threshold set too high misclassifies numerous dimensions as independent entities, devoid of any association with other factors. Hence, the calibration of a reasonable threshold is an exercise of paramount importance.

In practice, thresholds are typically determined by decision-makers

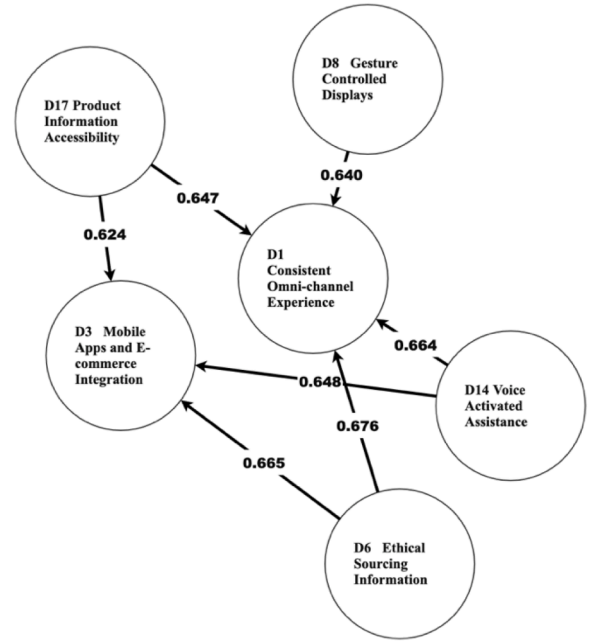

Fig. 2. Path of relation between dimensions.

or are the product of expert deliberation. Notably, various scholars have advanced distinct solutions to this conundrum. These range from the maximum mean difference entropy method Mukhametzyanov (2021) to the subjective method and the computation of the mean of the comprehensive influence matrix. The current paper assimilates elements from these approaches and settles upon a threshold set at the mean plus 2 standard deviations (0.640). With the discerning identification of key influencing factors, characterized by adherence to the mean plus 2 standard deviations threshold and in conjunction with the relationship pathways, the paper proceeds to construct a relationship path diagram encapsulating these pivotal factors. This graphical representation, prominently displayed in Fig. 3, serves as an indispensable tool for visualizing the intricate nexus between these key dimensions. Frequently, complex decision dilemmas arise wherein multiple dimensions either exert influence on other dimensions (grouped as causative elements) or are themselves influenced by other dimensions (categorized as effectual elements). The enhancement of any single dimension not inherently led to system-wide improvements due to the interdependence of these dimensions. Consequently, understanding the interplay among dimensions is crucial to identify those within the causative group that could be enhanced, thereby improving the dimensions within the effectual group and subsequently enhancing the entire system (Costa-jussà et al., 2022). With this in view, this study harnessed a fusion of fuzzy logic and DEMATEL techniques to delineate the causative and effectual connections among dimensions. To refine the analysis, a threshold value of 0.640 was established to exclude relatively insignificant effects.

The choice of the mean plus 2 standard deviations (0.640) as the threshold in this study stems from an examination of diverse factors geared towards augmenting the lucidity and dependability of the analytical findings. The mean plus 2 standard deviations threshold delicately balances the preservation of complexity with the avoidance of undue simplification in the cause-effect relationships among dimensions. Should the threshold be set too low, it risks precipitating an overwhelming convolution of relationships, thereby obstructing the interpretative clarity of the outcomes. Conversely, an excessively high threshold erroneously compartmentalize dimensions as independent entities, neglecting their actual interdependences. The selection of the mean plus 2 standard deviations threshold is methodically tailored to address these considerations. Integrating two standard deviations above

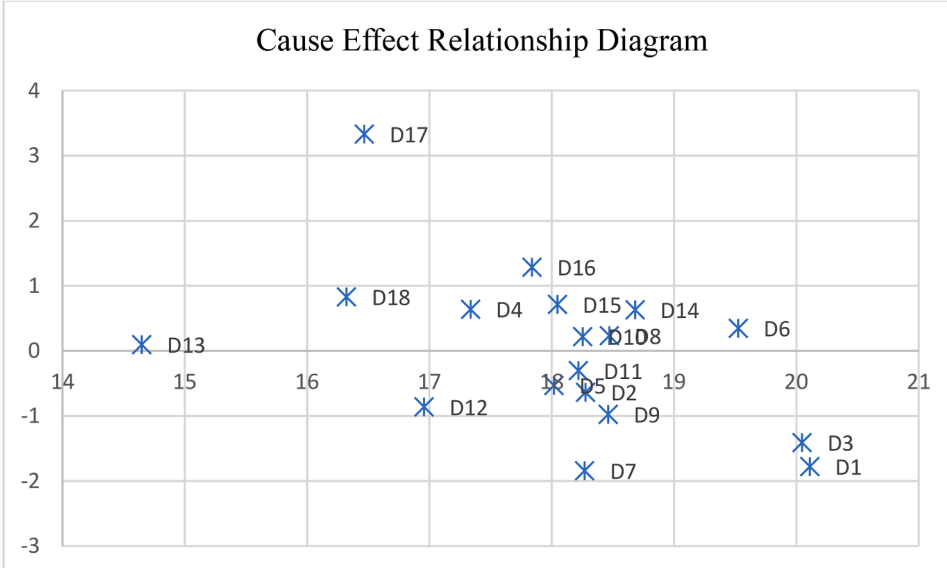

Fig. 3. Nexus between the dimensions.

the mean, the threshold encapsulates a substantial swath of data, spotlighting relationships that manifest a degree of consistency and significance surpassing chance variations. This statistical rigor imbues credibility to the identified key influencing factors, thereby bolstering the reliability of the outcomes. Furthermore, the mean plus 2 standard deviations threshold harmonizes with best practices in threshold determination, drawing inspiration from established methodologies like the maximum mean difference entropy and subjective approaches. This amalgamation of methodologies engenders a comprehensive yet nuanced approach to threshold establishment, enriching the methodological integrity of the study. The amalgamation of fuzzy logic and DEMATEL methodologies has significantly contributed to assisting managers in pinpointing key dimensions.

Predicated on the ($R_i + C_j$) values, the factors are ranked as follows: D1 > D3 > D6 > D14 > D8 > D9 > D2 > D7 > D10 > D11 > D15 > D5 > D16 > D4 > D12 > D17 > D18 > D13. According to this ranking, a consistent omni-channel experience (D1) emerges as the most pivotal catalyst for customer expectations in a retail store.

As the demarcations between retail and service industries blur, customer demands gradually shift from the product itself to the product and the service offered. A seamless omni-channel experience during the process of enhancing the purchase experience can mitigate risks, shorten service or product delivery cycles, enhance the novelty and market acceptance of services and products, ensuring the successful implementation of technology for green service consumption. The integration of mobile apps and e-commerce (D3) ranks as the second most critical enabler, a finding supported by prior literature. Mobile Apps and E-commerce play a pivotal role in accumulating purchase experience and knowledge, constituting important sources of creative value for future green consumption designs. Since they are in direct contact with customers, they can understand customer needs, deliver viable, creative solutions and align with customer expectations. The third most crucial enabler is Ethical Sourcing Information (D6). Retail stores need to prioritize adherence to ethical sourcing information and develop data protection and ethical usage protocols. This practice enhances communication with customers and judgment skills. It supports all stages of green consumption and customer expectations by facilitating the exchange of information, the design of new services, market supervision, introduction of new services to the market and control in information sourcing. Voice Activated Assistance (D14) and Gesture Controlled Displays (D8) also hold pivotal roles. Voice Activated Assistance is essential for promptly providing customers with the necessary aid and obtaining feedback for new product development. As customer expectations are highly personalized, retail stores often collaborate with external stakeholders to realize green service consumption.

To affect these service adaptations, retailers need to allocate additional resources and securing access to these resources necessitates top management's approval and support. Thus, support and integration are vital factors in driving the development of new services. The core of customer expectations lies in the transformation of unencoded knowledge into encoded knowledge. Activities such as Community Engagement Platforms and Product Information Accessibility promote this process of knowledge accumulation, transfer and creation, resulting in continuous growth. Combining Ri-Dj with Ri, it becomes evident that market volatility exerts the most significant influence, second only to Mobile Apps and E-commerce Integration. It can be inferred from this that retail stores play a substantial role in shaping customer expectations in green service consumption. Retailers can gain deeper insights into the dynamic competitive landscape, which aligns with customer expectations. Simultaneously, technological integration assists retailers in better grasping the value of new services, facilitating the more effective launch of green service consumption activities. Fig. 2 illustrates that Consistent Omni-channel Experience and Mobile Apps and E-commerce Integration are the most direct factors influencing customer expectations. The other factors subsequently influence customer expectations through their interaction with these two primary drivers.

6. Discussion

6.1. Theoretical Connection and Contribution to Literature

In the realm of information sciences, the is success model postulates that an enhancement in the quality of a system, service, or information directly correlates with heightened customer satisfaction (DeLone & McLean, 2003). This foundational premise suggests that the integration of advanced technology not only elevates the quality of service but also ensures the delivery of superior information and system performance. Consequently, this leads to increased customer engagement, driven by the satisfaction derived from the improved service quality. Within this theoretical framework, the model in the study posits that the integration of technology in retail environments functions similarly by enhancing the quality of service, which in turn fosters customer engagement. This engagement, underpinned by customer satisfaction, is crucial for encouraging sustainable consumer behaviours, such as green

consumption. Furthermore, the application of structuration theory provides an additional layer of understanding. According to this theory, the interaction between rules and resources shapes the social structure, producing outcomes that manifest as social practices (Iyamu, 2019). In the context of a retail store, the integration of technology can be seen as a means of refining these rules and optimising resource availability. The enhanced structure not only facilitates smoother operations but also nurtures a retail environment conducive to customer engagement. This engagement is essential as it drives green consumption, a social practice emerging from the interplay between technology-driven structural improvements and customer participation. By synthesising these theoretical perspectives, it becomes evident that the strategic deployment of technology in retail stores not only improves service quality but also influences the social structure, resulting in increased customer engagement and the promotion of green consumption practices. This theoretical extension underscores the critical role of technology in shaping sustainable consumer behaviour within the modern retail landscape.

The study makes substantial contributions to the literature, particularly within the domain of information systems, by extending existing knowledge on customer expectations in retail environments undergoing technological transformations. Firstly, the study enhances theoretical frameworks by integrating concepts from multiple disciplines, including economics, technology, and consumer behavior, to understand the dynamics of green service consumption. This interdisciplinary approach enriches the theoretical panorama, addressing the multifaceted nature of retail environments and customer expectations. Building on this theoretical integration, one significant methodological contribution is the application of the fuzzy DEMATEL method to analyze causal relationships among factors influencing customer expectations. This technique, previously used in various fields to unravel complex interrelationships (Lu et al., 2013; Kilic et al., 2020), is employed here to provide a nuanced understanding of how different dimensions such as in-store and online integration, sustainability practices, and transparency impact customer satisfaction. This application adds depth to the studies such as van der Heijden (2006), Lee (2010), Olson and Chae (2012) by demonstrating the utility of advanced decision support tools in retail research. Moreover, the research emphasizes the growing importance of green service consumption, a relatively nascent area within information system literature. Exploring how eco-friendly practices and ethical sourcing influence customer preferences, the study builds on previous work that highlighted the need for integrating sustainability into business practices (Zahay & Griffin, 2004; Galbreth & Ghosh, 2013; Touboulic et al., 2014). This focus aligns with the global trend towards sustainability and provides valuable insights for retailers aiming to balance ecological responsibilities with customer expectations.

6.2. Key Parameters Influencing Preferences and Satisfaction

In addition to theoretical and methodological advancements, the study offers practical guidance for retail businesses, highlighting actionable strategies for enhancing customer satisfaction through technological integration and sustainable practices. The identification of key drivers such as omnichannel experiences, mobile app integration, and the accessibility of ethical sourcing information helps retailers prioritize efforts and align their service offerings with customer expectations. These insights extend the work of previous studies that have explored the impact of technology and service innovation on customer engagement and satisfaction (Stickel, 1995; Pourjavad & Shahin, 2018). Acknowledging that customer expectations are dynamic and subject to change, the study suggests the need for continuous adaptation and responsiveness in retail strategies. This perspective builds on earlier research that examined the fluid nature of customer preferences in response to technological advancements and market trends (Wolf, 1995). The study's findings underscore the importance of ongoing research to further dissect the evolving roles of different dimensions in shaping customer expectations. Furthermore, the study emphasizes the importance of knowledge exchange and collaboration within business networks to improve the reliability of green service consumption. This insight is crucial for retailers operating in turbulent markets, as it highlights the need for robust decision-support tools and collaborative interactions with downstream partners. This contribution aligns with prior research advocating for enhanced knowledge sharing and collaboration to drive service innovation and meet customer expectations (Verma et al., 2021). Subsequently, the study significantly enriches the information system literature by providing a comprehensive analysis of the factors influencing customer expectations in technologically transformed retail environments, with a particular focus on green service consumption.

The study explores customer expectations in retail environments, undergoing technological transformations, with a particular focus on green service consumption. Analyzing various factors, the study identifies key parameters that influence customer preferences and satisfaction. These parameters span across different dimensions, including the seamless integration of in-store and online experiences, the implementation of sustainable and green practices, opportunities for customer engagement and interactivity and the transparency and accessibility of information. The intricate interplay between these dimensions highlights the study's interdisciplinary nature. It underscores the ongoing shift from a product-centric approach to a service-oriented model within the retail sector, fueled by technological advancements and the growing importance of green service consumption principles. The research has addressed the research questions (RQ1 and RQ 2), providing insights into the key parameters influencing customer expectations of technological transformations enhancing the purchase experience in retail stores, with a specific focus on green service consumption. The identified parameters encompass a range of dimensions, including in-store and online integration, sustainability and green practices, engagement and interactivity and transparency and information. These parameters collectively contribute to shaping consumer preferences and satisfaction in the retail environment. The interaction among these dimensions reveals a complex web of relationships, emphasizing the interdisciplinary nature of the study. The transition from a product-centric to a service-oriented economy, the role of technology as a transformative force and the incorporation of green service consumption principles underscores the multifaceted dynamics at play. Relationships among influencing factors, methodological insights and the dynamic nature of customer expectations further illuminate the intricate interplay that defines the retail sector.

In recent years, with a growing prevalence of retailers in national economies, the global economy has progressively transitioned from a product-centric model to a service-oriented one, driven by the imperative of meeting customer expectations (Mahdikhani, 2023). A mounting number of retail establishments have begun recognizing the pivotal role of services in generating value, seeking novel avenues for economic growth through service content innovation and the transformation of service delivery formats towards green service consumption (Verma et al., 2021). Nevertheless, the challenge of effectively implementing these changes continues to perplex most retailers. The study significantly contributes to the existing body of knowledge in information systems research in several ways. Employing the fuzzy DEMATEL method, the research delves deeper, revealing the causal relationships between these factors (Razavisousan & Joshi, 2022). This knowledge empowers retailers to prioritize their efforts and focus on the factors that have the most significant impact on customer satisfaction. Secondly, the research emphasizes the growing area of green service consumption within the information systems field. It provides valuable insights for retailers looking to integrate eco-friendly practices while simultaneously meeting customer expectations (Otim & Grover, 2006; Benitez-Amado & Walczuch, 2012). Finally, the study offers practical guidance for retail businesses through actionable management implications. Highlighting factors like omnichannel experiences, mobile app integration, and the

accessibility of ethical sourcing information, the research assists retailers in strategically aligning their service offerings with customer expectations in the context of green service consumption (Levenburg & Klein, 2006; Deng et al., 2015).

6.3. Strategic Implications for Integrating Sustainable Practices in Retail

Conducting an exhaustive review of the extant literature and engaging in in-depth retail research, this study has delineated 18 factors that exert influence over customer expectations in the realm of retail stores. Employing the fuzzy DEMATEL method, this study elucidates the intricate web of interrelationships among these factors. Given the interdependence and mutual influence inherent in these factors, this study assists retailers in disentangling causality, thereby discerning which factors act as drivers of other factors. Concentrating on the most influential determinants, retail establishments can adeptly align themselves with customer expectations. This study yields a host of significant implications for management. Firstly, it stands as the pioneer in identifying the drivers of customer expectations and elucidating the causal connections empowering retailers to make informed decisions concerning the most influential factors. In line with the findings, both consistent omni-channel experience and Mobile Apps and E-commerce integration serve as vital catalysts and the most potent sources of creativity. Retailers need to proactively undertake measures to attract customers and fulfil their expectations. It's noteworthy, however, that despite their high ranking, both these enablers fall within the causative group based on the DEMATEL analysis. This signifies that retailers should enhance the enablers that have a direct impact on customers in order to facilitate the implementation of green service consumption, thereby attracting customers to foster new service development. Secondly, the satisfaction of customer expectations entails adaptability in response to the evolving demands of channel partners. The DEMATEL analysis indicates that Product Information Accessibility and Pricing Transparency are pivotal determinants, with Ethical Sourcing Information emphasizing the timely and accurate dissemination of critical information.

Consequently, to realize successful service innovation, retailers need to enhance their technological integration efforts and seek to overcome market volatility. This approach enhances the collection of market intelligence and the sharing of external knowledge, thereby rendering the exchange of knowledge more advantageous. Thirdly, as per the results, the timeliness and accuracy of sharing knowledge emerges as the quintessential requirement to meet customer expectations. Retailers need to develop policies and programs to induce desirable forms of technological integration within the firm network, as knowledge exchange stands as a fundamental element of success. Collaborative partnerships within business networks can lead to improved reliability in green service consumption, as retailers immerse themselves in the inner workings of these networks to benefit from cooperative synergies. Lastly, Sustainable Product Information emerges as a potent dimension, prompting retailers to recognize the significance of customer expectations and green service consumption. The impact on collaboration is maximized when firms confront rapid shifts in products and technology. Conversely, firms operating in turbulent market landscapes necessitate more robust implementation of advanced decision support tools and intensive improvements in collaborative interactions with downstream partners. These initiatives are indispensable. The factors categorized within the causative group, delineated through the DEMATEL analysis, harbor profound ramifications for retailers endeavoring to instigate favorable shifts in customer expectations. These factors encompass a spectrum of dimensions including the omnichannel experience, mobile application integration, e-commerce assimilation, product information accessibility, pricing transparency, ethical sourcing disclosure and sustainable product detailing.

Retailers need to strategically allocate resources and undertake initiatives aimed at fortifying these dimensions to adeptly address the evolving demands of their clientele. For instance, prioritizing the seamless integration of omnichannel experiences and the amalgamation of mobile applications and e-commerce platforms enables retailers to cultivate environments conducive to creativity and innovation while adhering to principles of eco-friendly service consumption. Moreover, ensuring the accessibility and transparency of product information and pricing engenders trust and augments customer contentment. Ethical procurement practices and the provision of sustainable product information further underscore retailers' dedication to societal and environmental stewardship, resonating with the discerning preferences of a progressively conscientious consumer base. Astutely harnessing these causative factors, retailers not only satisfy prevailing customer expectations but also position themselves as vanguards in spearheading positive transformations toward more sustainable and ethical retail paradigms. This holistic strategy not only amplifies customer satisfaction and allegiance but also fortifies retailers' competitive standing within the ever-evolving retail milieu.

7. Implications and Conclusions

7.1. Theoretical contribution

The contributions traverse a vast array of dimensions and abstractions, collectively enriching the theoretical panorama in this intricate domain. The shift from a product-centric economic paradigm to a service-oriented milieu, as expounded, underscores the multidisciplinary essence of the study It accentuates that comprehending the evolving realm of green service consumption necessitates insights emanating from a spectrum of academic realms, encompassing but not limited to economics, technology and the intricacies of consumer behavior. This perspective accentuates the criticality of interdisciplinary scholarly exploration to fathom the intricate dynamics at play. These interconnections are pivotal in grasping how diverse factors engage with one another, mutually exert influence and collectively delineate the totality of the customer experience. The development of theoretical frameworks adept at modelling these intricate interrelationships assumes paramount importance, for they serve as conduits for invaluable insights into the labyrinthine dynamics that mould customer expectations. The introduction of causality within the retail ecosystem constitutes a seminal concept in this discussion. The exploration of causal linkages amongst the factors exerting influence adds a substantial layer of complexity to the theoretical framework. This analytical methodology empowers retailers and researchers alike in the pinpointing of primary drivers propelling customer expectations, thereby facilitating the formulation of more informed strategies and targeted interventions.

7.2. Practical Implication

The recurrent motif highlighting technology's pivotal role in shaping customer expectations resonates throughout the research and also articulating the aims of SDG 9. Whether through omni-channel experiences, mobile applications, or the seamless integration of e-commerce, technology emerges as a transformative fulcrum within the retail sector. This emphasis underscores the inexorable need for incessant technological innovation and progressive strides in delivering a seamless and gratifying retail experience. The discussion introduces the central theme of green service consumption, with a particular focus on the criticality of sustainable practices, ethically sourced products and eco-friendly strategies. These facets contribute to the broader narrative surrounding sustainability within the retail sphere. Their inclusion augments the theoretical comprehension of how retail can harmonise with ecological and ethical imperatives. It underscores the weightiness of comprehending consumer behaviour, preferences and expectations. The principle of aligning technological transformations with the express needs of the customer emerges as a foundational tenet, underscoring the primacy of a human-centric approach within the overarching theoretical

framework. The introduction of methodological instruments such as fuzzy logic and DEMATEL as instruments for scrutinising intricate relationships within the retail context bestow a richness to the tools available to researchers and practitioners exploring the realm of green service consumption. The study acknowledges that customer expectations are not static, they exhibit a state of flux in response to the dynamism of market forces and technological advancements. This recognition is pivotal for the sculpting of nuanced theoretical constructs that remain consonant with the ever-evolving terrain of green service consumption.

7.3. Societal inclusions

The shift from a product-centric economic paradigm to one centered on services, exemplifies the societal integration. This transition signifies a profound societal metamorphosis, wherein consumer desires and anticipations are in a state of flux. Customers are progressively attributing greater significance not solely to the product but to the orbit of services enveloping it. The societal trend towards valuing experiences and services holds paramount significance. The societal dimension extends further to the evolving landscape of sustainability. The burgeoning awareness and concern regarding environmental and ethical issues among the broader society are integral components of green service consumption. Consumers are becoming increasingly conscientious about the ecological and ethical ramifications of the products and services. This heightened societal consciousness molds the strategies and modus operandi of retailers in response to these expectations. Additionally, the societal perspective is pivotal in comprehending the causal relationships among influential factors. The infusion of technology into retail practices and customer interactions mirrors the societal integration of digital transformation. Consumers anticipate seamless digital engagements, such as online shopping, mobile applications and augmented reality experiences. The theme of green service consumption itself is inherently interwoven with societal values and expectations. The accentuation of sustainable practices, ethical sourcing and eco-friendly methodologies aligns with the societal emphasis on environmental preservation and ethical production. Retailers need to be responsive to these societal imperatives by integrating green practices into their services to meet the advancing expectations of environmentally and ethically aware consumers. Furthermore, the voice of the customer is central to sculpting the strategies of retailers. Comprehending and meeting the dynamic and evolving anticipations of customers constitutes a societal inclusion that propels retail innovations and service ameliorations.

8. Conclusion

While it's reasonable to study green service consumption and customer expectations as a cohesive whole, customer expectations exhibit dynamic characteristics. Consequently, while the identification of dimensions from a static viewpoint is comprehensive and methodical, it doesn't account for the differing roles these dimensions play at different stages. Future research should strive to dissect the dynamic process of customer expectations, discerning the variegated functions of these dimensions from a dynamic perspective. Secondly, the study has established rudimentary relationships among dimensions, albeit with certain limitations imposed by the subjectivity of the experts participated in the study. To elevate the significance and credibility of these findings, forthcoming research should leverage structural equation modeling to validate the pathways connecting the influencing factors. This pledge to continuous scholarly inquiry ensures the continual progression and adaptability of the field in response to the evolving demands of green service consumption. In summary, the study delves into green service consumption by exploring customer expectations on technological transformations enhancing purchase experience in retail store. It underscores the pivotal shift from a product-centric to a service-oriented economy, emphasizing the evolving nature of customer expectations. The theme of green service consumption is central, highlighting the importance of sustainable practices and ethical sourcing driven by societal values. The exploration of causal connections among influencing factors reveals the significant impact of societal forces on the retail sector. Technology's indispensable role, a customer-centric approach and methodological insights further enrich the discussion. The dynamic nature of customer expectations is recognized and the discourse concludes by acknowledging its limitations and advocating for future research. These insights collectively advance the theoretical landscape surrounding technological transformations in green service-focused retail experiences.

CRediT authorship contribution statement

Arjun J Nair: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Sridhar Manohar:** Writing – review & editing, Validation, Project administration, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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